M-Tech 1st Semester

MID-TERM SEM 1 MATHEMATICAL FOUNDATION OF COMPUTER SCIENCE

FULL MARKS 30 CSC - 901 **TIME 1 HR**

1. Answer any 5 Questions. (5X2=10)

a. Two cards are drawn from the pack of 52 cards. Find the probability that both are diamonds or both are kings.

b. A problem is given to three persons P, Q, R whose respective chances of solving it are 2/7, 4/7, 4/9 respectively. What is the probability that the problem is solved?

c. Suppose that the cdf of X is given by:

$$F(a) = \begin{cases} 0 & \text{for } a < 0 \\ \frac{1}{2} & \text{for } 0 \leq a < 2 \\ \frac{3}{4} & \text{for } 2 \leq a < 4 \\ 1 & \text{for } a \geq 4. \end{cases}$$

d. Class has 50 students: 20 male (M), 25 brown-eyed (B)
For a randomly chosen student, what is the range of possible values for p = P (M U B)?

e. Lucky Lucy has a coin that you're quite sure is not fair.
- They will flip the coin twice
- It's your job to say whether it more probable that the

Tosses are the same, i.e. HH or TT, or different, i.e. HT or TH.
Which should you choose?
i. More probable they are the same
ii. More probable they are different
iii. Doesn't matter
iv. It depends on the actual probabilities of getting heads or tails.

f. Suppose that X takes values between 0 and 1 and has probability density function 2x. Compute Var(X).

g. Suppose that P(A) = 0.4, P(B) = 0.3 and $P((A \cup B)^C) = 0.42$. Are A and B independent?

2. Answer any two questions (10X2=20)

A. i) Suppose you are taking a multiple-choice test with c choices for each question. In answering a question on this test, the probability that you know the answer is p. If you don't know the answer, you choose one at random. What is the probability that you knew the answer to a question, given that you answered it correctly? (7)

ii) Suppose the next patient enrols and their age is 97 years.
How does the mean and median change?
To get the median, order the data: 21, 24, 34, 34, 42, 44, 46, 52, 56, 64, 97
If the age were recorded incorrectly as 977 instead of 97, what would the new median be? What would the new mean be?(3)

B. If birth weights in a population are normally distributed with a mean of 109 oz and a standard deviation of 13 oz,
i. What is the chance of obtaining a birth weight of 141 oz or heavier when sampling birth records at random? (5)
ii. What is the chance of obtaining a birth weight of 120 or lighter? (5)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

C. i) The teacher recorded the following test scores for her class: [85, 78, 92, 88, 84, 90, 79, 81, 87, 93, 78, 84]. (6)
Calculate the mean score.
Determine the median score.
Identify the mode(s) if any.

ii) Find covariance for following data x = {6,5,3,4,2}, y = {12,10,8,6,4} (4)

END

M. Tech, Semester I, Mid Semester Examination I, Total marks: 30 Duration: 1 hour

1. Answer any *5* questions from the following: [2 x 5= 10]

a) Show the recursive steps of multiplying 1234 and 5678 using the D& C strategy that runs in $O(n^{\log 3})$ time
b) Can a binary tree that is not full correspond to an optimal prefix code? Justify your answer
c) What is the total number of parenthesization possible for a sequence of 5 matrices A_1, A_2,A_3,A_4, A_5?
d) Show that the lower bound on any comparison based sorting is $\Omega(n \log n)$
e) Explain how optimality is ensured in Greedy and Dynamic Programming approaches to solve a problem.
f) Compute the optimal parenthesization of matrix chain multiplication for the matrix sequence <A1 A2 A3> having dimension 10 x 20, 20 x 5, 5x 30 respectively for A1, A2, A3
g) If the n element input of Select algorithm is already sorted, and the algorithm is used to find the kth smallest element, how many recursions will be required to get the kth smallest element?

2. Answer any *2* questions from the following: [10 x 2= 20]

a) i) Give the Dynamic Programming formulation for the all pair shortest path problem.

ii) A contiguous subsequence of a list S is a subsequence made up of consecutive elements of S. For example if S is 5,15,-30,10,-5,40,10, then 15,-30,10 is contiguous subsequence but 5,15,40 is not. You are given a list of numbers $a_1, a_2, \ldots, a_n$, formulate a dynamic programming formulation to compute a contiguous subsequence whose sum is maximum. For example, 10, -5, 40, 10 with sum 55 is maximum.
[5+5]

b) i) Let g denote the size of each group in 'Select' Algorithm for some positive integer g>= 3. Derive the running time of the algorithm in terms of g.

ii) Let A[1:n] and B[1:n] be two arrays of distinct integers sorted in increasing order. Give an efficient algorithm to find the median of the 2n elements in both A and B. What is the running time of your algorithm?
[5+5]

c) i) What is an optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers? a:1 b:1 c:2 d:3 e:5 f:8 g:13 h:21

ii) Generalize your answer to find the optimal code when the frequencies are the first n Fibonacci numbers?

[7+3]

University of Calcutta

1st Midterm M.Tech. Semester- I Examination, 2025

Subject: Computer Science & Engineering

Paper Code & Name: CSE901 & MACHINE LEARNING

Full Marks: 30 **Time: 1:00 Hour**

1. Answer any five of the following questions: [2x5=10]

i. What are the different forms of learning? Define machine learning..

ii. How is the programming flow of machine learning different from regular programming? What are other forms of programming?

iii. How does the individual error of any labelled data affect the cost function of any machine learning algorithm? Briefly explain your answer.

iv. What are log-scaling and normalization of any dataset, and what are their implications?

v. What is 'critical word first' and how does it reduce miss penalty?

vi. Compare and contrast deductive and inductive learning.

2. i. What assumptions must be satisfied before finding a suitable linear regression model for a given labelled dataset? How can you justify the sustainability of the model?

ii. What are the different charts/graphs used to preprocess the datasets? What are their significances? Briefly discuss.

iii. Explain the concept of overfitting in the context of linear regression. How can it be prevented or mitigated?

iv. Consider the dataset: x=[1,2,3,4,5], y=[2,3,5,7,11]

(a) Fit the simple linear regression $y=\beta_0+\beta_1 x$ by least squares.

(b) Compute SSE and R^2

(c) Interpret β_1. [5x4=20]

M.Tech 1st Semester–(Mid-Semester)Examination 2025

Subject:ComputerScience

Paper Code & Name: CSE903 (Wireless &Mobile Computing)

Date: 14.09.2025 **FullMarks:30** **Time:12.30PM– 1:45PM**

Answer question no 1 and any two from the rest

1. Answer any five questions[5x2=10]

   a. What is the typical radiation pattern of a half-wave dipole antenna, and why is it important for Wi-Fi systems?
   b. What is the significance of ISM Band in wireless communication?
   c. "TDMA in combination with FDMA is used in GSM while CSMA/CA is used in Wi Fi networks" – why?
   d. State the specific contribution(s) of CSMA MAC protocols over S-Aloha.
   e. State a situation where the contention free MAC protocol may be a preferred one than contention based protocols.
   f. "Flooding is a well known broadcasting technique" – State the process and highlight the principal advantage and disadvantage of the said process.
   g. State the reason behind proposing exponential back off time in CSMA/CD.

2. a) Describe the hidden station problem.
   b) How is the situation tackled in W-LAN?
   c) Why is the PCF maintained in WLAN; even if the provision of DCF is in place? [3+5+2 =10]

3. a) Compare the performance of Distance vector and Link State routing in terms accuracy and bandwidth requirement with proper justifications.
   b) State and derive the mathematical expression for the two-ray path loss expression. [5+5=10]

4. a) Why do we need frequency reuse in cellular networks Define co-channelinterference in the context of frequency reuse. Derive its formula in terms of Cluster Size and path loss exponent.
   b) If the system bandwidth B=200 kHz, required MDR=1 Mbps, path loss exponent n=4, calculate the minimum cluster size for frequency reuse.

[(1+1+4)+4=10